World Journal of Acupuncture – Moxibustion 28 (2018) 231–236

Contents lists available at ScienceDirect

World Journal of Acupuncture – Moxibustion

journal homepage: www.elsevier.com/locate/wjam

Clinical Research

The effect of electroacupuncture preconditioning on cognitive impairments following knee replacement among elderly: A randomized controlled trial ☆

Fei-yi ZHAO (赵非一)[a], Zhe-yuan ZHANG (张浙元)[b], Ying-xia ZHAO (赵英侠)[c], Hai-xia YAN (燕海霞)[c], Yu-fang HONG (洪钰芳)[a], Xiao-jie XIA (夏小芥)[d], Hong XU (许红)[a,*]

[a] *Shanghai Municipal Hospital of TCM Affiliated to Shanghai University of TCM, Shanghai 200071, China (上海中医药大学附属市中医医院, 上海200071, 中国)*
[b] *Haiyan People's Hospital, Zhejiang 314300, China (海盐人民医院, 浙江314300, 中国)*
[c] *School of Basic Medical Sciences, Shanghai University of TCM, Shanghai 201203, China (上海中医药大学基础医学院, 上海201203, 中国)*
[d] *Department of Education, East China Normal University, Shanghai 200062, China (华东师范大学教育学部, 上海200062, 中国)*

ARTICLE INFO

Article history:
Available online 30 November 2018

Keywords:
Postoperative cognitive dysfunction
Electroacupuncture
Preconditioning
Knee replacement
Elderly

ABSTRACT

Objective: To investigate if electroacupuncture (EA) preconditioning can mitigate cognitive impairments and reduce the incidence of postoperative cognitive dysfunction (POCD) following knee replacement and its safety among elderly.
Methods: Totally 60 participants met the inclusion criteria were enrolled in a randomized controlled trial a ratio of 1:1, with 30 cases allocated to the treatment group and 30 cases allocated to the control group, respectively. The participants in the treatment group were provided with real-EA therapy whereas participants in control group were provided with placebo-EA therapy (Streitberger Placebo-needle). In both groups, *Tou sanshen* (头三神) acupoints, including Sìshéncōng (四神聪EX-HN1), Shéntíng (神庭 GV24), and bilateral Běnshén (本神GB13) were adopted as the main acupoints, while Bǎihuì (百会GV20), bilateral Hégǔ (合谷LI4), and bilateral Tàichōng (太冲LR3) were adopted as matching acupoints. Interventions were offered 5 days prior to the surgery, once daily, and continued for total 5 days. The global scores of Mini-Mental State Examination (MMSE), and levels of serum inflammatory cytokines including interleukin 1β (IL-1β) and tumor necrosis factor-α (TNF-α), and S100-β protein were observed at 24 h prior to the surgery, and postoperative 24 and 72 h respectively for assessing the incidence of POCD and the severity of cognitive impairments among patients. Meanwhile, adverse effects were monitored and recorded.
Results: (1) Compared with baseline, MMSE global scores in both treatment and control groups markedly decreased at postoperative 24 h. MMSE global scores in treatment group decreased from 29.43 ± 0.97 to 27.10 ± 1.95 while that in control group decreased from 29.27 ± 1.01 to 26.83 ± 2.25 (all ?$P < 0.05$), and this trend continued until postoperative 72 h (at postoperative 72 h, MMSE global scores in treatment group was 26.53 ± 2.26 versus 24.79 ± 3.03 in control group). Moreover, decline in control group was more significant than that in treatment group at postoperative 72 h ($P < 0.05$). (2) Compared with baseline, levels of serum IL-1β, TNF-α and S100-β in both groups increased markedly at postoperative 24 and 72 h. IL-1β in treatment group increased from 43.13 ± 5.51 to 73.13 ± 2.32 at postoperative 24 h and reached 83.17 ± 5.95 at postoperative 72 h, while IL-1β in control group increased from 44.87 ± 5.83 to 91.10 ± 3.55 at postoperative 24 h and reached 111.93 ± 9.18 at postoperative 72 h; TNF-α in treatment group increased from 51.27 ± 6.48 to 88.80 ± 3.55 at postoperative 24 h and reached 94.37 ± 5.22 at postoperative 72 h, while TNF-α in control group increased from 52.07 ± 7.48 to 116.37 ± 3.14 at postoperative 24 h and reached 121.40 ± 3.68 at postoperative 72 h (both ?$P < 0.05$), furthermore, increases of IL-1β and TNF-α levels in control group were more significant ($P < 0.05$). Statistical difference in level of S100-β was not observed ($P > 0.05$). (3) There was no statistical difference in POCD incidence at postoperative 24 h

☆ Supported by Key and Weak Discipline Construction Project (Gerontology of TCM), Shanghai Municipal Commission of Health and Family Planning (2015ZB050), Project of Science and Technology Commission of Shanghai Municipality (16401902600).

* Corresponding author.
E-mail address: 0449@szy.sh.cn (H. XU).

https://doi.org/10.1016/j.wjam.2018.11.003

and postoperative 72 h between two groups ($P > 0.05$), though the incidence of POCD in patients receiving real-EA therapy was indeed much lower than that in patients receiving placebo-EA therapy, particularly at postoperative 72 h (POCD incidence rate at postoperative 24 h in treatment group was 26.67%, 30.00% in control group; POCD incidence rate at postoperative 72 h in treatment group was 30.00%, 46.67% in control group). (4) No serious adverse events were reported in this trial. No one dropped out from this trial.

Conclusion: EA preconditioning can mitigate cognitive impairments at post-knee replacement surgery 24 and 72 h in elderly through inhibiting expression of inflammation. However, there is insufficient evidence to support that EA pretreatment can reduce the incidence of POCD.

Introduction

As a global public health issue in the field of perioperative care, and with a high incidence ranging from 25% to 40% among elderly patients [1], postoperative cognitive dysfunction (POCD), is defined as a common neurological complication following prolonged and major surgery. Though POCD and postoperative delirium (POD) are often discussed together because both of them are brain dysfunction following surgery with high morbidity [2], difference between POCD and POD might not be neglected. Distinguished from POD featuring acute change or fluctuation in consciousness and perception, and patients with POD are often disoriented due to the decreased awareness of outside environment [3], POCD is characterized by deficits in memory, attention, concentration, and language comprehension, or even progressive personality changes in serious circumstances but patients with POCD are oriented [1,3–6].

Many studies emphasize that due to the significant decline in patient's self-care ability and the heavy burden on both individual and the whole society, more attention should be paid to various impacts of POCD on the long-term life quality of patient. To illustrate, the influence of POCD on patient's postoperative life could be assessed from several aspects, including the patient's survival period, labor force status, and dependence on medical security scheme and health care system [7].

Though the definite pathogenesis of POCD remains unclear, it is usually identified as an interaction of a variety of factors [8]. According to laboratory examination, agminated amyloid beta peptide (Aβ) similar to the typical feature of Alzheimer's disease is also found in patient with long-term POCD. The level of preoperative plasma Aβ1-42 and Aβ1-40 are closely link to the incidence of POCD as well [9]. Anesthesia might be responsible for the development of POCD as well. Evidence from pharmacological studies reveal that since different drugs can act on different cells and molecules within various brain regions, the choice of anesthetic agents thereby may predict the incidence and prognosis of POCD [10]. Inflammation response seems to play a pivotal role in POCD. In term of existing reports, both systemic- and neuro-inflammation triggered by anesthesia or surgical traumas are involved in the development of POCD [6,11]. Hence, there are speculations that inhibition of inflammatory expression might be an option in preventing or treating POCD, though more evidence resulting from larger sample size animal experiments and clinical trials is needed [12]. It is worth mentioning that the relationship between endothelial cells and POCD has been paid increasing attention recently. According to previous studies, many products activated by platelets and endothelial cells were also reported as key biological markers in evaluating POCD status [13].

Angiotensin II receptor antagonists such as candesartan, α-adrenoceptor agonists such as dexmedetomidine, and free radical scavengers such as edaravone are currently used for prevent POCD [14]. However, potential risk of hypotension and bradycardia induced by dexmedetomidine [15], angioedema and sacroiliac joint pain induced by candesartan [16], and headache induced by edaravone [17] may limit the use of these medications, which prompts patients to seek novel approaches to manage POCD. Acupuncture, as a key component of complementary and alternative medicine, is widely used in Asia and increasingly in Western countries, and showed satisfactory effects in improving cognitive deficits caused by a variety of diseases [18].

Through reviewing previous literature, some clinical trials about electroacupuncture (EA) on POCD have been reported. However, since no placebo-EA were set as effective control in the trial [19,20], decreased reliability and validity of the results seemed inevitable. To provide more credible evidence of EA on POCD, particularly investigate if EA pretreatment can reduce the incidence of POCD following knee replacement surgery and its safety, in this study, a randomized, single-blind, placebo-controlled clinical trial with more intuitive and reliable detection techniques was conducted. In addition to setting placebo-EA as control group, both subjective assessment scale such as Mini-Mental State Examination (MMSE), and objective laboratory indicators such as content of serum interleukin 1 beta (IL-1β), tumor necrosis factor-α (TNF-α) and S100β protein were introduced in this trial in order to comprehensively evaluate the decline of postoperative cognitive function among elderly patients, and address the design defects in relevant previous studies. Meanwhile, adverse effects were monitored and recorded as well.

Clinical data

Participants

Sixty elderly patients aged 60–75 years old, American Society of Anesthesiologists ranging from grade I to II (ASA I means a normal healthy patient; ASA II means a patient with mild systemic disease [21]), undergoing knee replacement surgery at either Shanghai Municipal Hospital of TCM affiliated to Shanghai University of TCM or Haiyan People's Hospital affiliated to Zhejiang Pharmaceutical College from March 2015 to August 2018 were selected. This study was approved by the ethics committees at both two hospitals mentioned above (Ethical No. 2015SHL-KY-17/2017SHL-KY-05). All participants were required to sign informed consent before participating in the trial.

Inclusion criteria

(1) 60–75 years old; (2) Patients without cognitive dysfunction (MMSE $\geq$ 27 points); (3) Did not take any psychoactive drugs within 4 weeks prior to the commencement of this trial; (4) Signed informed consent, voluntarily participated in this clinical trial, and pledged to cooperate with follow-up visits.

Exclusion criteria

(1) Those with cancer, severe hepatic and renal insufficiency, hematopoietic system and endocrine system primary diseases;

(2) Patients with cognitive impairments caused by neurodegenerative diseases such as Parkinson disease, Alzheimer's disease, Vascular Dementia, Stroke or other psychiatric disorders, and other systemic chronic illness and medical condition (MMSE $<$ 27 points); (3) Hamilton Anxiety Scale $\geq$ 14 points, and (or) Hamilton Depression Scale $\geq$ 18 points; (4) Pre-estimation of operation duration $>$ 4 h or intraoperative blood loss $>$ 1000 ml; (5) Those participated in other clinical trials within the past 1 month; (6) Those with a tendency of skin infections and bleeding; (7) Patients who were deaf, blind and had disorders in communication.

Withdrawal and dropout criteria

(1) A serious adverse event related to the research occurred; (2) When a participant asked to withdraw from the trial for any reason at any time; (3) The treatment had to be interrupted due to the unforeseen reasons; (4) Participant accepted other treatments during this trial or didn't cooperate with the researcher.

Methods

Participants and randomization

This was a double-center, randomized, patient-blinded, parallel-group and placebo-controlled trial with a 1:1 ratio to treatment and control group. Before the commencement of recruitment, the random numbers were generated by SPSS 21.0 statistical software, and these random sealed numbered were placed into 60 opaque envelopes. A total of 60 patients met the inclusion criteria were recruited by hospital-based advertisements from the outpatient clinic and wards of orthopedic department in Shanghai Municipal Hospital and Haiyan People's Hospital. Those eligible patients were randomly allocated to the different groups through simple envelope method.

Intervention methods and courses

All patients received routine perioperative care: (1) observing the recovery of the affected limb, including peripheral blood supply, skin temperature, sensation, and pulsation of the dorsal pedal artery; (2) hemostasis by compression bandage, and indwelling drainage tube; (3) ambulatory electrocardiographic monitoring. Real-EA or placebo-EA preconditioning were provided five days consecutively prior to the surgery.

(1) treatment group (Real-EA)

Acupoints: Sìshéncōng (四神聪EX-HN1), Shéntíng (神庭 GV24), Bǎihuì (百会GV20), bilateral Běnshén (本神GB13), bilateral Hégǔ (合谷LI4), and bilateral Tàichōng (太冲LR3).

Method: Participants in treatment group were in dorsal position for needling acupoints at both head and extremities. Acupoints skin was sterilized by 75% alcohol. Standard sterilized disposable needles (0.25 mm in diameter and 25 mm in length, stainless steels, Jiajian Medical Instrument Co., Ltd. China) were inserted horizontally into the acupoints of EX-HN1, GV20, GV24 and GB13, while LI4 and LR3 were inserted perpendicularly with another specification, standard acupuncture needles (0.25 mm in diameter and 40 mm in length, Jiajian Medical Instrument Co.,Ltd. China). Acupuncture depths were various according to different acupoints. Usually, needles were inserted into the skin to a depth of 10–30 mm due to the fatty tissues at the site of the acupoints and manipulated manually until '*deqi*' (an irradiating feeling considered to be indicative of effective needling) achieved. The feelings of '*deqi*' were strengthened by acupuncturist through twisting or thrusting the needles. After that, all of the needles were retained for further 30 min before removal. In addition, the needles inserted into LI4 and LR3 were connected to the EA device (Model: G6805-2; Shanghai Huayi Medicinal Instruments Co., Ltd. China) with dilatational wave, 2/100 HZ, 3 mA.

(2) control group (Placebo-EA)

Acupoints: Acupoints were chosen as same as those in treatment group.

Method: Placebo acupuncture approach (Streitberger Placebo-needle) [22] was selected in the control group. Streitberger needle was a type of specially designed acupuncture needle with a blunt needlepoint. When the blunt needlepoint touched the skin, it would spring back into the needle sheath. Similar to real-EA group, the needles positioned at LI4 and LR3 were connected to the same model but modified EA device, but the device was not activated except the indicator light was flashing, so that the patients mistakenly believed that the EA device was working.

Patients in both groups were treated once daily for total five consecutive days. Meanwhile, they were informed that these two acupuncture approaches were commonly used in clinical practice and both of them were effective, this trial was just to compare which acupuncture approach was much better for reducing the incidence of POCD. Patients' other questions about the treatment efficacy during the intervention period would not be answered by the acupuncturists.

(3) anesthesia method

Both groups were given the same method of general anesthesia. When general anesthesia was performed, intravenous Fentanyl (2 μg/kg), Midazolam (0.04 mg/kg), Propofol (1 mg/kg), and Cisatracurium Besilate for Injection (0.2 mg/kg) were administered sequentially. Mask oxygen inhalation and artificially assisted breathing were provided as well. Endotracheal intubation was performed after the patient' muscles were relaxed, anesthesia machine was connected, and then mechanical ventilation and end-tidal carbon dioxide partial pressure monitoring were given. Intraoperative anesthesia was maintained by constant infusion of Remifentanil 0.2–0.5 μg/(kg min) and Propofol 4–6 mg/(kg h) to the end of surgery. During the surgery, blood pressure and heart rate were maintained within ±20% of the baseline value, and bispectral index was maintained at 40–60. After surgery, endotracheal tube was pulled out until the patient was awake.

Outcome measures

Outcome assessments were conducted at 24 h before the surgery, and postoperative 24 and 72 h, respectively. The concentrations of serum S-100β, IL-1β and TNF-α were tested by enzyme-linked immunosorbent assay method.

(1) primary outcome

MMSE: Mini-Mental State Examination (MMSE)is a commonly used, useful and economical clinical assessment tool to screen for cognitive impairment [23]. Originally designed by Folstein et al. in 1975, MMSE includes the following seven domains: orientation to time, orientation to place, registration, attention and calculation, recall, language, repetition, and complex commands. Each domain contains at least 1 question, 1 point for each question, and total 30 questions. According to the global scores: 27–30 points indicates interviewee without cognitive dysfunction; 21–26 points indicates interviewee with mild cognitive dysfunction; 10–20 points indicates interviewee with moderate cognitive dysfunction; 0–9 indicates interviewee with severe cognitive dysfunction. POCD could be diagnosed if postoperative MMSE score is at least 2 points lower than preoperative MMSE score [24].

Table 1
Baseline balance test of POCD participants' demographic characteristics.

Groups	Patients	Age (years, Mean ± SD)	Gender n(%)		Marital status n(%)		
			Male	Female	Never married	Married/cohabiting	Divorced/widowed
Treatment	30	65.23 ± 4.03	12 (40)	18 (60)	1 (3.33)	28 (93.33)	1 (3.33)
Control	30	66.70 ± 3.84	14 (46.67)	16 (53.33)	0 (0.00)	27 (90)	3 (10)

Table 2
Baseline balance test of POCD participants' clinical characteristics (Mean ± SD).

Group	Patients	MMSE global scores	Serum marker (μg L^{-1})		
			IL-1β	TNF-α	S-100β
Treatment	30	29.43 ± 0.97	43.13 ± 5.51	51.27 ± 6.48	0.17 ± 0.03
Control	30	29.27 ± 1.01	44.87 ± 5.83	52.07 ± 7.48	0.18 ± 0.04

(2) secondary outcomes

Serum IL-1β and TNF-α are main inflammatory cytokines produced by microglia, and can objectively show the level of systemic inflammatory response. Like inflammatory cytokines, serum S100β which is toxic to neurons and is closely related to POCD, is another credible bio-marker for assessing POCD status.

Statistical analysis

SPSS 21.0 statistical software was used for statistical description and inference. Measurement data in normal distribution were expressed as Mean ± Standard Deviation (SD) and analyzed by paired t-test or two dependent samples t-test for inter-group comparisons and intra-group comparisons. The measurement data in abnormal distribution or ranked data were analyzed by Wilcoxon rank-sum test. The enumerations data were expressed by ratio and processed by Chi-square (X^2) test. Repeated Measure ANOVA will be used to investigate differences between two groups on most endpoints except those abnormal distribution data or data with heterogeneity of variance will be tested by Kruskal–Wallis H after variable transformation. P-value < 0.05 were considered as a significance level in statistical analysis.

Results

Demographic and clinical characteristics

After baseline assessment, demographic and clinical characteristics of the participants were presented (Tables 1 and 2). As can be seen, baseline clinical characteristics were balanced between the two groups (All $P > 0.05$). Meanwhile, there were no significant differences were depicted in demographic characteristics between two groups as well.

Cognitive functions and inflammation levels

The results of MMSE global scores related to changes of cognitive function, and levels of serum IL-1β, TNF-α and S-100β related to changes of systematic inflammation status are presented in Table 3. In term of changes in MMSE global scores and inflammation markers, significant decline in cognitive function associated with increased release of inflammatory factors following the surgery can be observed in both groups. According to Table 3, compared with baseline, MMSE global scores in both treatment and control groups markedly decreased at postoperative 24 h (both $P < 0.05$), and this trend continued until postoperative 72 h. Moreover, decline in control group was more significant than that in treatment group at postoperative 72 h ($P < 0.05$). It might be concluded that cognitive impairments occurred in both groups following the surgery. Furthermore, the cognitive dysfunction of patients who received placebo-EA were more severe than that of patients who received real-EA preconditioning. Compared with baseline, levels of serum IL-1β, TNFα, and S-100β in both groups increased markedly at postoperative 24 h (All $P < 0.05$), and this trend continued until postoperative 72 h. Compared with treatment group, levels of serum IL-1β and TNFα increased more significantly in control group at postoperative 24 and 72 h (All $P < 0.05$). Regardless at postoperative 24 or 72 h, S-100β level in treatment group was lower than that in control group, though the difference did not meet the statistical standard (both $P > 0.05$).

Incidence of POCD at postoperative 24 and 72 h (Table 4)

Among 60 patients, 17 were diagnosed with POCD at post-knee replacement surgery 24 h, with an incidence of 28.33%. Of these, 8 were from the treatment group and 9 were from the control group. The incidence of POCD was 26.67% in treatment group versus 30.00% in control group. At postoperative 72 h, 23 subjects were diagnosed with POCD, with an incidence of 38.33%. Of these, 9 were from the treatment group and 14 were from the control group. The incidence of POCD was 30.00% in treatment group versus 46.67% in control group. It suggested that the incidence of POCD might increase as time goes by within 3 days after surgery. In addition, although there was no statistical difference in POCD incidence at postoperative 24 h and postoperative 72 h between two groups ($P > 0.05$), the incidence of POCD in patients receiving real-EA therapy was indeed much lower than that in patients receiving placebo-EA therapy, particularly at postoperative 72 h.

Safety assessment

No serious adverse events were reported in this trial. No one dropped out from this trial.

Discussion

POCD is a common neurological complication following surgery among elderly patients. Epidemiological survey showed that, due to a variety of risk factors involved in the major surgery and the whole perioperative period, an average of 14% patients experienced cognitive decline and mental confusion within a few weeks to three months after surgery [25]. Furthermore, both cardiosurgery and specific orthopedic procedures are interventions with a relatively high incidence of POCD [6]. Though POCD is usually reversible, there are some patients who will experience delays in rehabilitation and a low life quality because of POCD. The rate of disability and mortality might also increase within one year after surgery, which leads to a huge burden on both families and society. In addition, POCD may also accelerate the progression of

Table 3
Comparison of MMSE global scores, concentration of serum IL-1β, TNF-α and S-100β between two groups of POCD patients at 24 h before surgery, postoperative 24 and 72 h (Mean ± SD).

Group	Patients	Assessment endpoint	MMSE global score	Serum marker (μg L^{-1})		
				IL-1β	TNF-α	S-100β
Treatment	30	Baseline	29.43 ± 0.97	43.13 ± 5.51	51.27 ± 6.48	0.17 ± 0.03
		Postoperative 24 h	27.10 ± 1.95^{a}	73.13 ± 2.32^{ab}	88.80 ± 3.55^{ab}	0.38 ± 0.34^{a}
		Postoperative 72 h	26.53 ± 2.26^{ab}	83.17 ± 5.95^{ab}	94.37 ± 5.22^{ab}	0.28 ± 0.20^{a}
Control	30	Baseline	29.27 ± 1.01	44.87 ± 5.83	52.07 ± 7.48	0.18 ± 0.04
		Postoperative 24 h	26.83 ± 2.25^{a}	91.10 ± 3.55^{a}	116.37 ± 3.14^{a}	0.49 ± 0.42^{a}
		Postoperative 72 h	24.79 ± 3.03^{a}	111.93 ± 9.18^{a}	121.40 ± 3.68^{a}	0.41 ± 0.30^{a}

Notes: compared with baseline, $^{a}P < 0.05$; compared with control group, $^{b}P < 0.05$

Table 4
Incidence of POCD at postoperative 24 and 72 h n(%).

Groups	Patients	Postoperative 24 h	Postoperative 72 h
Treatment	30	8 (26.67)	9 (30.00)
Control	30	9 (30.00)	14 (46.67)
X^2-*Value*		0.082	1.763
P		0.774	0.184

Alzheimer's disease [26,27]. Acupuncture is a characteristic therapy of Traditional Chinese Medicine (TCM). Due to its advantages, including wide adaptability, safety and low cost, acupuncture is popular in China and gradually being accepted by western countries. Previous studies have illustrated that acupuncture could play a role in brain protection through improving brain function, significantly reducing neurological deficits, and then promoting learning, memory, and other cognitive functions [28,29]. Hence, a randomized, single-blind, placebo-controlled clinical trial was conducted in our study, in order to investigate if EA preconditioning can reduce the incidence of POCD among elderly patients undergoing knee replacement surgery. The safety of the therapy was monitored at the same time.

In light of the results, the decreased MMSE global scores and increased levels of serum IL-1β, TNFα, and S-100β after surgery showed that cognitive function in elderly patients would be significantly impaired due to surgical trauma or usage of anesthetic agents in knee replacement surgery. Furthermore, this kind of cognitive impairments seemed to increase as time goes by at least within 7 day.

Though the cognitive impairments seemed to be inevitable, EA preconditioning was proven to effectively alleviate this kind of damages, which might be reflected by the changes of serum IL-1β, TNFα, and S-100β levels. As can be seen in Table 3, higher levels of IL-1β, TNF-α and S-100β were observed in control group at postoperative 24 and 72 h. Although the exact and definite pathogenesis of POCD is still unclear, evidence is accumulating for a series of inflammatory responses [30,31]. It can be concluded from this study that the protective effect of EA preconditioning on cognitive function might be achieved by inhibiting inflammation during or after the surgery. It should be noted that the increased trend of S-100β was mitigated undergoing acupuncture pretreatment though it was not significant between two groups ($P > 0.05$). However, this result was not enough to show that EA is ineffective in regulating S-100β. Three potential reasons for this result were concluded as follow: (1) sample size was not large enough; (2) stimulation time of EA or the observation period was too short; (3) S-100β might not be sensitive enough for testing mild cognitive damages. In addition, no statistical difference in comparison of incidence of POCD between two groups ($P > 0.05$), though the number of POCD patients who received real-EA preconditioning was less than that of those who received placebo-EA preconditioning. The finding suggested that there was not enough evidence to support the efficacy of EA pretreatment on reduction of incidence of POCD.

In this study, EX-HN1, GV24, and GB13 were selected as main acupoints combined with GV20, LI4, and LR3 as matching acupoints. GV24 is the convergent acupoint of governor meridian, bladder meridian and stomach meridian. Since governor meridian links three *yang* meridians of hand and three *yang* meridians of foot to brain, GV24 can be viewed as a gathering place of *qi* and blood in governor meridian. Meanwhile, bladder meridian is derived from the mind and brain, and administrates the activities of mind and brain. Therefore, *qi* and blood in meridians will be regulated and delivered to brain through acupuncturing GV24. EX-HN1 is extraordinary acupoint with an effect in improving brain health and promoting intelligence as well. GB13 is the convergent acupoint of gallbladder meridian and Yang link vessel/Yang wei meridian, with the effect of tranquilizing and allaying excitement. In addition, GV20 is not only passed by governor meridian, but also is the convergence of various meridian. Regulation of *yin* and *yang* in *zang-fu* might be achieved by acupuncturing GV20 as well. Besides, *Siguan* (fore gates四关) acupoints consisting of bilateral LI4 and LR3, which are commonly used for tranquilizing and sedating the mind with a satisfactory curative effect in the treatment of neurologic and mental diseases. Animal study [32] showed that EA at *Siguan* acupoints could improve the learning and memory ability of Alzheimer's disease model rats through improving excitability of cerebral cortical, promoting cerebral circulation, and regulating signal transduction pathways. Previous clinical trial [33] also found that specific brain functional regions of elderly could be regulated by acupuncturing *Siguan* through activating frontal lobe and posterior cingulate cortex.

In terms of no adverse events reported in this trial, and no participants withdrew or dropped out from this trial, we believed that five consecutive days of EA preconditioning might be acceptable and safe for elderly patient who is going to undergo knee replacement surgery. Based on the results of present trial, EA preconditioning is worthy of clinical promotion in reducing the incidence of POCD.

Limitations

Both strengths and some methodological limitations were available in this trial, including the observation period is short, which may result in the statistical bias in outcome assessments. According to previous studies [23], POCD will occur at any time within 3 months following the surgery. Therefore, the ideal observation and outcome assessment period should be set ranging from the end of the surgery to 3 months posterior to the surgery. In addition, the bias may also be caused due to the insufficient sample size. For further research, conducting a multi-center, randomized, single-blind, placebo-controlled clinical trial with a larger sample and long-term observation of effects of acupuncture is warranted. Prospective, other laboratory examination indicators related to POCD such as interleukin-6, B-cell lymphoma 2, Bcl-associated x protein and Neuron specific enolase are also expected to be included as the object outcome, combined with self-reported outcome such as MMSE to provide more practical and stricter clinical research modalities in POCD clinical studies.

Conclusion

This trial provides evidence for the satisfactory short-term effects of EA preconditioning in mitigating cognitive impairments among elderly following knee replacement surgery, suggesting that EA therapy may be a potential, safe and useful non-pharmacologic intervention option for POCD management though more evidence-based medicine evidence is still required. However, there is insufficient evidence to support that EA pretreatment could reduce incidence of POCD.

Conflict of interest

The authors declared that there was no potential conflict of interest in this article.

Statement of informed consent

Informed consents were obtained from all individual participants in this study.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.wjam.2018.11.003.

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